Retrieval Augmented Generation Based Study Aid for Personalized Learning

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Abstract

In this project, I developed a Retrieval-Augmented Generation (RAG) system designed to generate personalized study guides for graduate students enrolled in the MIDS program at UC Berkeley taking Statistics 203 - Statistics for Data Science. My system retrieves and summarizes key concepts from both lecture units and textbook chapters, tailoring explanations to the student's preferred level of depth and clarity. The goal is to support students in reviewing complex topics such as probability, regression, or causal inference using contextually relevant and semantically rich language. I explore the impact of different chunk sizes on retrieval quality and experiment with various large language models (LLMs), including GPT-4o-mini, LLaMA 3, and Mixtral, to evaluate answer quality using BLEU, F1, and cosine similarity. Results demonstrate that model choice and chunk size significantly influence performance, with notable differences in how models handle semantic recall versus surface-level similarity.

Keywords : Retrieval-Augmented Generation, RAG, Study Guide Generation, Semantic Similarity, Large Language Models, BLEU, ROUGE F1, Cosine Similarity

Introduction

Graduate-level data science programs often challenge students with high volumes of technical material spread across multiple sources. Students are expected to synthesize content from lecture slides, textbook chapters, and assignments into a cohesive understanding. However, review processes are often manual, time-consuming, and mismatched with individual student learning preferences. To address this challenge, I propose a RAG-based system that generates tailored study guides by summarizing key content from a graduate-level statistics course in the MIDS program.Retrieval-Augmented Generation (RAG), which pairs a large language model with a document retriever, has proven effective in domain-specific applications such as university support chatbots (*Kuratomi et al., 2024; Antico et al., 2024*) and online course tutoring (*Lang & Gürpinar, 2024*).

This system builds on recent advances in retrieval-augmented generation, where a language model is paired with a document retriever to ground responses in specific source material. By adapting this framework to the education domain, I aim to create a system that not only retrieves accurate information but also presents it in a way that is cognitively accessible, and responsive to user input.Prior research has emphasized RAG's ability to balance factual grounding with user preference (Henkel et al., 2024), as well as its scalability for assisting students across disciplines (*Frontiers in Psychology, 2024; Gao et al., 2023*).

Background

Although large language models (LLMs) demonstrate strong text generation capabilities, they struggle in scenarios requiring access to structured knowledge bases or specific documents, limiting their effectiveness in knowledge-intensive tasks. To address this limitation, retrieval-augmented generation (RAG) models have been developed, enabling generative models to incorporate relevant document fragments into their inputs.

RAG-based systems offer a compelling solution for online learners and students navigating dense academic material, such as textbook chapters or research papers. Unlike general-purpose tools like ChatGPT, which struggle to ingest and retain course-specific documents in bulk, RAG frameworks can ingest full course materials, stay up to date with content changes, and surface contextually grounded answers. This capability is particularly valuable for exam preparation and for students in asynchronous programs who may lack access to live tutoring or collaborative study groups. Previous research has explored RAG for general QA and summarization, but limited work has applied it to educational review.While universities like UNC Chapel Hill have begun exploring RAG-driven tutors, publicly available implementations remain scarce, highlighting the need for tools like the one proposed in this project.

Approach

Data

The dataset consists of 14 lecture units and 7 textbook chapters from the UC Berkeley MIDS Statistics for Data Science curriculum. Each unit focuses on a core topic such as probability theory, regression, or causal inference. The source files were primarily in PDF format and were parsed using the PyPDFLoader library in LangChain, then preprocessed to extract structured text. Each document was tagged with metadata such as unit topics, textbook topics, and topic labels for easier filtering and analysis.

For retrieval purposes, documents were split into overlapping chunks ranging from 1000 to 8000 tokens using a recursive text splitter from LangChain. Ground-truth answers were generated using a keyword-guided chunk extraction algorithm, ensuring alignment with real instructional content. The final dataset includes over 1,000 meaningful chunks across all documents, providing rich context for RAG-based retrieval and generation.

For the prompting process, 2–3 prompt questions were manually written for each unit, ranging from easy to complex to simulate realistic study guide questions. In total,36 prompts were created across all 14 units. For example:

- **Unit 04**: "What is a conditional expectation?" (easy), "Derive the regression anatomy formula" (complex).

For each prompt, a corresponding gold chunk was extracted using a keyword-based search strategy, ensuring alignment with authoritative content from the course materials.These gold chunks were used as ground truth during evaluation.

This design prioritizes depth of evaluation and alignment with instructional content over large-scale quantity and allows for controlled benchmarking across model architectures and prompt difficulty.

Baseline

As a baseline, we used the GPT-4o-mini model from OpenAI with a fixed chunk size of 1000 tokens across all 14 units.This setup involved no chunk size tuning or model switching, allowing us to establish a controlled reference point for evaluating retrieval and generation quality.

We evaluated the baseline performance using 3 metrics BLEU,ROUGE F1 and cosine similarity.These metrics were selected based on their application in prior RAG literature,including the RAG-based Institutional Assistant project by Kuratomi et al. (2024), which used F1, cosine similarity (via MPNet embeddings), and lexical overlap to comprehensively assess answer quality.

The baseline configuration achieved average scores of BLEU = 0.014214, F1 = 0.106112, and Cosine = 0.389806, providing a strong semantic alignment and lexical match. These results indicate modest lexical overlap but fairly strong semantic alignment, especially in cases involving paraphrased or conceptually grounded responses.

Modeling

I experimented with three instruction-tuned large language models: GPT-4o-mini, Mixtral-8x7B-Instruct-v0.1, and LLaMA-3.3-3B-Instruct. Each model was tested with three different chunk sizes 2000, 4000, and 8000 tokens respectively.

All 3 models were integrated into a RAG pipeline using the LangChain framework.We used a similarity based retriever (ChromaDB) to fetch relevant chunks of text.Chunks were created using recursive text splitting with token sizes of 2000, 4000, and 8000 and a constant chunk overlap of 200. The retriever was configured to a similarity score threshold of 0.3 to filter out irrelevant chunks and the top 20 most relevant segments to limit the number of retrieved results. This allowed the model to focus on high-confidence, top-ranked evidence from the document collection while avoiding noise from loosely related contexts.The final answer was generated using a prompt template that incorporated both the retrieved context and the question.Each response was evaluated using BLEU (n-gram overlap) , Rouge F1 (recall and precision of key phrase) and cosine similarity (MPNet embeddings).

To evaluate model outputs against high-quality references, I adopted a gold chunk extraction approach inspired by methods in retrieval-augmented evaluation research. Specifically, for each prompt, I defined a set of target keywords corresponding to core instructional concepts (e.g, "conditional expectation," or "hypothesis test power") and extracted the most relevant document chunks containing these terms. These gold chunks served as proxy ground-truths for evaluating lexical overlap and semantic alignment, similar to how "nugget-based" reference answers are used in QA evaluation (Baumel et al., 2021). This approach allowed for consistent and curriculum-aligned benchmarking, even in the absence of human-labeled ground-truths.

Results

The results below show the average BLEU , ROUGE F1 and cosine similarity scores for the 3 models across 3 different chunk sizes :

Chunk Size	Model	BLEU (mean)	F1(mean)	Cosine(mean)
2000	gpt-4o-mini	0.012987	0.105531	0.386300
4000	gpt-4o-mini	0.010594	0.093167	0.376054
8000	gpt-4o-mini	0.012345	0.099329	0.388565

Chunk Size	Model	BLEU (mean)	F1(mean)	Cosine(mean)
2000	Mixtral	0.016388	0.111486	0.371172
4000	Mixtral	0.016671	0.112229	0.371383
8000	Mixtral	0.016662	0.112338	0.372172

Chunk Size	Model	BLEU (mean)	F1 (mean)	Cosine (mean)
2000	Llama3	0.017791	0.101273	0.383609
4000	LLama3	0.017791	0.101273	0.383609
8000	LLama3	0.017791	0.101273	0.383609

Mixtral model consistently delivered the highest ROUGE F1 scores, peaking at 0.1123 at a chunk size of 8000. This suggests the model's strength in preserving key phrases from the reference answers. BLEU scores for the Mixtral model remained stable across chunk sizes indicating consistent n-gram overlap. Cosine similarity hovered around 0.371–0.372, reflecting solid but slightly lower semantic alignment compared to GPT-4o-mini.

GPT-4o-mini model achieved the highest cosine similarity, with a peak of 0.3886 at chunk size 8000, indicating strong semantic alignment and effective generalization in cases involving paraphrased or abstract queries. Although its ROUGE F1 and BLEU scores were lower than Mixtral’s (ROUGE peaking at 0.1055 and BLEU at 0.0129), the model remained strong in capturing conceptual meaning.

LLaMA 3 produced the most consistent scores across chunk sizes, with identical values for each metric at all sizes (BLEU = 0.0178, ROUGE F1 = 0.1013, cosine = 0.3836). While its ROUGE F1 was lower than Mixtral’s, and cosine slightly below GPT-4o-mini, it outperformed both in BLEU, suggesting a tendency to preserve surface-level phrasing. The uniformity across chunk sizes also points to reduced sensitivity to input granularity.

Compared to the baseline configuration , all models demonstrated comparable or improved BLEU scores with Llama3 outperforming baseline in surface-level overlap and Mixtral outperforming it in ROUGE-F1.

Conclusion

Overall, the study demonstrates that both model architecture and chunk size meaningfully affect the quality of responses in a RAG-based educational assistant.The Mixtral model is more effective in capturing exact terminology, while GPT-4o-mini excels in delivering semantically aligned answers. These results affirm the importance of choosing a model based on the desired balance between lexical fidelity and conceptual understanding in educational applications. Chunk size also plays a role, with larger chunks (4000–8000) tokens yielding more robust outputs without overwhelming memory constraints.They generally improve semantic recall, though gains in precision (BLEU or ROUGE) vary by model.

These findings validate the feasibility of using RAG for building personalized study assistants.In the future, such a system could support students in real time , offering adaptive explanations,surfacing relevant course material on demand and even simulating personalized tutoring experiences. Future iterations could incorporate user feedback loops, integrate multiple course sources dynamically, and provide multimodal support for exam review and discussion-based learning.

Appendix

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